
CS 207: Applied Databases Practicum

Course Code	CS 207
Course Name	Applied Databases Practicum
L-T-P-C	0-0-3-2
Prerequisites	IC 150 Computation for Engineers
Intended for	UG
Distribution	Compulsory for CSE; CS elective for EE and ME
Semester	3rd
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Contents

A few lab lectures (8 hours spread over the semester):

- Architectures of DB applications: Client-server; UI-Business logic-DBMS; Browser UI-Web server-Business logic-DBMS.
- Introduction to SQL
- Introduction to E-R modelling and MVC
- Transactions what, how and when?
- Introduction to a scripting language, eg PHP, Python
- Introduction to HTML and especially HTML5
- Introduction to Javascript and Ajax/Comet
- Introduction to NoSQL

Lab assignments (listed below) require 3 hours in the lab, preceded by at least 3 hours at home. The weekly assignments are stage-wise demonstrations of the evolution of a mini-project stages are as follows:

- Week 1-2 Choice of mini-project a useful web-based tool.
- Week 3-4 Designing the data model and table schemas, testing the tables manually.
- Week 5-6 Design the UI flow the user view.
- Week 7 Creating the GUI forms and reports.
- Week 8-9 Putting together the Web UI flow with the appropriate data access.
- Week 10 Basic tool ready with full functionality
- Week 11 Improving UI using javascript and HTML5 features.
- Week 12 Using AJAX for better user-interaction.
- Week 13 Substituting portions of the data model using NoSQL databases

Text Books

- This course will use web-resources to cover course topics.